

# A resolution-free proof of the Duistermaat–van der Kallen nonvanishing theorem

Christopher D. Long\*

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## Abstract

We present a resolution-free semialgebraic proof of the following theorem of Duistermaat and van der Kallen: if the Newton polytope of a nonzero complex Laurent polynomial contains the origin, then infinitely many of its positive powers have nonzero constant term. The principal estimate is a uniform middle-dimensional sublevel bound for the restriction of a holomorphic volume form to a semialgebraic family. Polynomial differentiation and a finite-multiplicity projection estimate prove this bound; a dyadic decomposition then gives absolute integrability of the logarithmic volume form on every middle-dimensional semialgebraic chain on which the Laurent polynomial is bounded. The local residue construction of Duistermaat and van der Kallen identifies the constant-term generating function with a fiber period. Our estimate gives a primitive with a power-law endpoint bound, contradicting the simple pole forced by vanishing of all positive constant terms. Infinite nonvanishing follows by applying this minimal nonvanishing statement to powers of the polynomial. The standard semialgebraic inputs are stated explicitly. Global cycle transport is obtained by an explicit generalized-critical-value ODE rather than Whitney–Thom theory; the latter is retained as an alternative in an appendix.

## 1 Introduction and statements

Let  $d \geq 0$  and

$$R_d = \mathbb{C}[x_1^{\pm 1}, \dots, x_d^{\pm 1}].$$

For  $0 \neq f = \sum_{\alpha \in \mathbb{Z}^d} a_\alpha x^\alpha \in R_d$ , define

$$A = \text{supp}(f) = \{\alpha : a_\alpha \neq 0\}, \quad \text{Newt}(f) = \text{conv}(A), \quad c_n(f) = \text{CT}(f^n) \quad (n \geq 0).$$

In particular  $c_0(f) = 1$ . For  $d = 0$ , the ring is  $\mathbb{C}$  and the exponent space is the zero-dimensional vector space.

**Theorem 1.1** (Duistermaat–van der Kallen, nonvanishing). *Let  $d \geq 0$  and  $0 \neq f \in R_d$ . If  $0 \in \text{Newt}(f)$ , then for every integer  $M \geq 0$  there is an integer  $n > M$  such that  $c_n(f) \neq 0$ .*

**Theorem 1.2** (Minimal nonvanishing). *Let  $d \geq 0$  and  $0 \neq f \in R_d$ . If  $0 \in \text{Newt}(f)$ , there exists  $n \geq 1$  such that  $c_n(f) \neq 0$ .*

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\*Email: galizur@gmail.com.

Theorem 1.1 follows algebraically from Theorem 1.2, by applying it to  $f^{M+1}$ ; see Section 9. Thus only the minimal statement needs an analytic proof. Strict separation subsequently gives

$$(\text{CT}(f^n) = 0 \text{ for every } n \geq 1) \implies \text{CT}(hf^n) = 0 \text{ for all sufficiently large } n$$

for every fixed Laurent polynomial  $h$ . The corresponding assertion for finite Fourier sums on a compact torus follows by integrating characters.

The original proof is due to Duistermaat and van der Kallen [3]. Kiselman's Question 9.1, recording a question of Passare, asks for a simpler proof of infinite nonvanishing [12]. The argument here is organized around a direct estimate for a restricted complex differential form. Its local residue construction and primitive-differentiation identity come from [3, Sections 4–5]; the new argument to be examined is the sublevel estimate and its use to control chains that may escape to infinity. We make no claim of bibliographic priority for individual auxiliary lemmas.

## Proof architecture and scope

After the elementary face reduction, assume that  $d \geq 1$  and  $0 \in \text{int Newt}(f)$ . Put

$$\Omega_d = dz_1 \wedge \cdots \wedge dz_d, \quad \omega_d = \frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_d}{x_d}.$$

The argument has the following implications:

uniform finite-multiplicity projection estimate

$$\begin{aligned} \implies & \int_{E \cap \{|P| \leq \varepsilon\}} |\Omega_d| \leq C\varepsilon^{1/D} \\ \implies & \int_{S \cap \{|f| \leq R\}} |\omega_d| \leq C_{S,f} R^{1/D} \\ \implies & \left| \int_a^{2a} J(s) ds \right| \longrightarrow 0 \quad (a \downarrow 0). \end{aligned}$$

If all positive constant terms vanish, the residue identity forces  $J(s) = -(2\pi i)^{d-1}/s$ . Its integral over  $[a, 2a]$  is a fixed nonzero constant, giving a contradiction.

The term *resolution-free* is literal: no resolution of singularities is used. The global transport step in Section 6 works directly on the closed affine embedding  $(\mathbb{C}^\times)^d \simeq \{(z, w) : z_j w_j = 1\} \subset \mathbb{C}^{2d}$ . Generalized critical values give a finite exceptional set, and a minimal-norm right inverse of  $df$  gives an explicit horizontal ODE. Its linear growth in the proper affine radius yields complete path lifting by Gronwall. Hardt triviality is retained for semialgebraic control, in particular for the final real sweep near zero. Appendix A records the older projective-graph/Whitney–Thom transport as an independent alternative; it is not a dependency of the main proof. The presentation remains relative to the explicitly stated standard semialgebraic inputs rather than reconstructing their foundations. The stronger critical-value and exponential-growth statement in [3, Theorem 5] is not a theorem of this draft.

## 2 Elementary Newton-polytope reductions

**Lemma 2.1** (Minimal-face reduction). *Suppose  $0 \in \text{Newt}(f)$ . There is a Laurent polynomial  $\tilde{f}$  in  $r \leq d$  variables such that*

$$\text{CT}(\tilde{f}^n) = \text{CT}(f^n) \quad (n \geq 0),$$

*and either  $r = 0$  and  $\tilde{f}$  is a nonzero constant, or  $r \geq 1$  and  $0 \in \text{int Newt}(\tilde{f})$  in  $\mathbb{R}^r$ .*

*Proof.* Let  $Q$  be the smallest face of  $\text{Newt}(f)$  containing 0. Then  $0 \in \text{relint } Q$ . If  $Q$  is proper, choose an exposing linear functional  $\ell$  with  $\ell = 0$  on  $Q$  and  $\ell > 0$  on  $\text{Newt}(f) \setminus Q$ . This is possible because the minimum occurs at  $0 \in Q$ . If  $Q$  is the whole polytope, use  $\ell = 0$ .

Let  $f_Q$  consist of the terms of  $f$  with exponent in  $Q$ . A sum of  $n$  support exponents can equal 0 only if every summand belongs to  $Q$ : apply  $\ell$  and use nonnegativity. Comparing the full coefficient expansions, including all coefficients and cancellations, gives  $\text{CT}(f^n) = \text{CT}(f_Q^n)$ .

The affine span  $L$  of  $Q$  is linear because  $0 \in Q$ . Since  $Q$  is a lattice polytope,  $L$  is rational and  $\Lambda = L \cap \mathbb{Z}^d$  is a lattice of rank  $r = \dim L$ . Choose a  $\mathbb{Z}$ -basis of  $\Lambda$  and express  $f_Q$  in the corresponding group algebra  $\mathbb{C}[\Lambda] \simeq R_r$ . The zero exponent and all products are preserved. If  $r = 0$ , its sole coefficient is nonzero. Otherwise the image of  $Q$  has 0 in its full-dimensional interior.  $\square$

**Lemma 2.2** (Powers of Newton polytopes). *For  $0 \neq f \in R_d$  and  $q \geq 1$ ,*

$$\text{Newt}(f^q) = q \text{Newt}(f).$$

*Proof.* Every exponent of  $f^q$  is a sum of  $q$  exponents of  $f$ , giving one inclusion. Let  $v$  be a vertex of  $\text{Newt}(f)$ , and choose a linear functional uniquely maximized at  $v$  on the support. If  $\alpha_1 + \dots + \alpha_q = qv$ , equality at this maximum forces  $\alpha_1 = \dots = \alpha_q = v$ . Thus the coefficient at  $qv$  is  $a_v^q \neq 0$ . All vertices of  $q \text{Newt}(f)$  occur in  $f^q$ , proving the opposite inclusion. The singleton case is immediate.  $\square$

**Lemma 2.3** (A unimodular chart at a vertex). *Suppose  $d \geq 1$  and  $0 \in \text{int Newt}(f)$ . After a unimodular monomial change of variables, which preserves every constant term of every power, one can write*

$$f(y) = y_1^{-m_1} \dots y_d^{-m_d} u(y), \quad m_i \in \mathbb{Z}_{>0}, \quad u \in \mathbb{C}[y_1, \dots, y_d], \quad u(0) \neq 0. \quad (2.1)$$

*Proof.* Choose a vertex  $v$  of  $\text{Newt}(f)$  and let  $C$  be the open cone of real linear functionals whose minimum on the support occurs uniquely at  $v$ . Choose a primitive lattice vector  $a_1 \in C$ , and extend it to a lattice basis  $a_1, b_2, \dots, b_d$  of the dual exponent lattice. Such a primitive vector exists by rational density in the nonempty open cone, followed by division by the gcd of its coordinates.

For sufficiently large integers  $N_i$ , the vectors  $a_i = b_i + N_i a_1$  lie in  $C$ : divide by  $N_i$  and use openness. These vectors together with  $a_1$  still form a lattice basis. In the resulting exponent coordinates every support exponent  $\alpha$  satisfies  $\langle a_i, \alpha \rangle \geq \langle a_i, v \rangle$ , with simultaneous equality only at  $v$ . Since 0 is an interior point, every nonzero linear functional takes negative values on the support; hence  $\langle a_i, v \rangle < 0$ . Set  $m_i = -\langle a_i, v \rangle$ . Factoring out  $y^{-m}$  gives (2.1) with constant coefficient  $a_v$ .

The exponent change is a lattice automorphism, so it preserves zero exponents, products, and constant terms. From this point one may regard  $f$  as the transformed Laurent polynomial and use the standard ordered form in the  $y$ -coordinates; no unrecorded orientation sign is then introduced.  $\square$

### 3 Semialgebraic integration and uniform projections

Semialgebraic means semialgebraic over  $\mathbb{R}$ , with arbitrary real parameters allowed. Complex coordinates are identified with their real and imaginary parts. Dimension in this section is real dimension.

#### 3.1 The standard inputs

The following precise forms of standard results will be used.

**Standard input 3.1** (Semialgebraic decomposition and triviality). A semialgebraic set and a finite collection of semialgebraic subsets admit compatible finite decompositions into Nash manifolds. Semialgebraic families have uniform bounds on the number of connected components of their fibers. For a continuous semialgebraic map  $g : X \rightarrow Y$ , there is a finite semialgebraic partition  $Y = \coprod_j Y_j$  and semialgebraic homeomorphisms

$$g^{-1}(Y_j) \simeq Y_j \times F_j$$

commuting with projection to  $Y_j$ . Compatible triangulations may be chosen for compact semialgebraic sets. Compact singular homology classes on semialgebraic manifolds admit finite semialgebraic cycle representatives.

These are the decomposition, triangulation, and Hardt triviality results in [1, 7]; the local-triviality statement is [1, Theorem 9.3.2]. A noncompact set need not have a finite triangulation by *closed* simplices. Only compact cycle representatives and compact truncated chains will require that assertion here.

**Standard input 3.2** (Integration and Stokes). Integration of a smooth form over a finite semialgebraic chain is compatible with finite subdivision and semialgebraic reparametrization. A bounded semialgebraic set of dimension  $k$  has finite  $k$ -dimensional Hausdorff measure. For a compact semialgebraic chain  $C$  and a smooth  $(k-1)$ -form  $\eta$  on a neighborhood of its support,

$$\int_C d\eta = \int_{\partial C} \eta.$$

On a smooth semialgebraic manifold, integration of a closed form over semialgebraic cycles agrees with the usual homology pairing.

The integration convention and change of variables are described in [6, Sections 1–2]; Stokes is [6, Theorem 2.9], extended to finite chains by subdivision and cancellation of common faces. Compatible  $C^1$  triangulations give another foundation [14]. The finite-volume assertion can also be obtained by summing the coordinate-projection bounds proved below. We use ordinary change of variables on smooth pieces, not a slicing theorem for currents.

### 3.2 The absolute density of a complex form

**Definition 3.3.** Let  $S \subset \mathbb{R}^N$  be semialgebraic of dimension at most  $k$ , and let  $\xi$  be a smooth complex-valued  $k$ -form on a neighborhood of  $S$ . On a smooth  $k$ -dimensional stratum  $M$ , let  $\tau_M(x)$  be either unit simple  $k$ -vector spanning  $T_x M$ . Define

$$\int_S |\xi| = \sum_{\dim M=k} \int_M |\xi_x(\tau_M(x))| d\mathcal{H}^k(x). \quad (3.1)$$

Lower-dimensional strata contribute zero. For  $k=0$  this is counting measure. For a finite chain  $C = \sum_j n_j C_j$ , the positive quantity used for estimates is  $\sum_j |n_j| \int_{C_j} |\xi|$ ; we write it as  $\int_{|C|} |\xi|$ . The oriented integral is linear in the chain.

The definition is independent of the chosen smooth decomposition, by common refinement and the change-of-variables formula. It satisfies

$$|g\xi| = |g| |\xi|, \quad |\xi + \zeta| \leq |\xi| + |\zeta|, \quad \left| \int_C \xi \right| \leq \int_{|C|} |\xi|. \quad (3.2)$$

The first identity concerns the modulus of the *full* restricted complex form. Expanding into real coordinate wedges is permissible for upper bounds, not as an alternative definition.

For example, on  $E = \{(z, -z) : 1 \leq |z| \leq 2\}$  one has  $(dz_1 \wedge dz_2)|_E = 0$ , although the sum of the absolute values of its separate real coordinate terms is nonzero. This cancellation will be essential in Section 4.

### 3.3 The uniform projection estimate

**Lemma 3.4** (Uniform finite-multiplicity integration). *Let  $\Theta$  be semialgebraic and let  $\mathcal{E} \subset \Theta \times \mathbb{R}^N$  be a semialgebraic family with  $\dim E_\theta \leq k$ . Suppose  $h_\theta : E_\theta \rightarrow \mathbb{R}^k$  is a semialgebraic family of continuous maps, differentiable on the smooth pieces after decomposition. There is  $N_0 < \infty$ , depending on this entire family, such that*

$$\int_{E_\theta} |dh_{\theta,1} \wedge \cdots \wedge dh_{\theta,k}| \leq N_0 \operatorname{vol}_k(h_\theta(E_\theta)) \quad (\theta \in \Theta). \quad (3.3)$$

*An unbounded image is allowed; the right side may be infinite.*

*Proof.* Retain the parameter in the map

$$H : \mathcal{E} \longrightarrow \Theta \times \mathbb{R}^k, \quad H(\theta, x) = (\theta, h_\theta(x)).$$

By Standard input 3.1, the numbers of connected components of all its fibers are bounded by one integer  $N_0$ . In particular the cardinality of every zero-dimensional fiber is at most  $N_0$ .

Fix  $\theta$ . Decompose  $E_\theta$  into smooth pieces on which the rank of  $h_\theta$  is constant. Pieces of dimension less than  $k$ , and pieces on which the rank is less than  $k$ , have zero pullback  $k$ -density. On rank- $k$  pieces of dimension  $k$ , the map is locally a diffeomorphism. The set of target points whose whole fiber has positive dimension has dimension at most  $k - 1$ ; its rank- $k$  inverse images also have measure zero, by local change of variables. Away from this exceptional set there are at most  $N_0$  preimages in total. Apply ordinary change of variables on injective smooth pieces, or equivalently the area formula with this counting multiplicity, and sum to obtain (3.3). This is the parameter-uniform version of [6, Proposition 2.5]; the parameter-retaining device also appears in the proof of its Proposition 2.8.  $\square$

*Remark 3.5.* In applications,  $\theta$  includes all varying set parameters, polynomial coefficients, and cutoffs. Parameters need not range in a compact set. Uniformity follows from one fixed semialgebraic description, not from compactness of parameter space. Adding finitely many prescribed semialgebraic inequalities produces another fixed family, to which the same lemma applies with a possibly different constant.

**Corollary 3.6.** *For a semialgebraic family  $E_\theta$  of dimension at most  $d$  contained in a fixed bounded box in  $\mathbb{C}^d$ ,*

$$\sup_\theta \int_{E_\theta} |\Omega_d| < \infty.$$

*More generally, the Euclidean  $d$ -volumes of a bounded semialgebraic family of subsets of  $\mathbb{R}^N$  of dimension at most  $d$  are uniformly bounded.*

*Proof.* Expand  $\Omega_d$  into its  $2^d$  real coordinate wedge terms and use (3.2) and Lemma 3.4 for the corresponding coordinate projections. Their images lie in fixed boxes. For the Euclidean-volume assertion, the coordinate components of a unit simple  $d$ -vector have squares summing to 1, so their absolute values have sum at least 1. Sum over all coordinate  $d$ -projections.  $\square$

## 4 A middle-dimensional polynomial sublevel estimate

Fix  $d \geq 1$ ,  $L \geq 1$ , a semialgebraic family  $E_\theta$  of real dimension at most  $d$  in the box  $[-L, L]^{2d} \subset \mathbb{C}^d$ , and  $D \geq 1$ . For a polynomial  $P$ , write

$$\|P\|_{\text{coef}} = \max_{\alpha} |[z^\alpha]P|.$$

**Theorem 4.1** (Uniform sublevel estimate). *There is  $C = C(\mathcal{E}, d, D, L)$  such that every holomorphic polynomial  $P \in \mathbb{C}[z_1, \dots, z_d]$  with  $\deg P \leq D$  and  $\|P\|_{\text{coef}} = 1$  satisfies*

$$\int_{E_\theta \cap \{|P| \leq \varepsilon\}} |\Omega_d| \leq C\varepsilon^{1/D} \quad (\theta \in \Theta, \varepsilon > 0). \quad (4.1)$$

The same estimate, with value zero on the left, holds for  $\varepsilon = 0$ .

**Lemma 4.2** (The one-derivative estimate). *For each  $i \in \{1, \dots, d\}$ , let*

$$\Xi_i(P) = dP \wedge dz_1 \wedge \dots \wedge \widehat{dz_i} \wedge \dots \wedge dz_d.$$

*There is  $C_1$ , uniform in  $\theta, P, \varepsilon, r$  and  $i$ , such that*

$$\int_{E_\theta \cap \{|P| \leq \varepsilon, |\partial_i P| \geq r\}} |\Xi_i(P)| \leq C_1 \varepsilon \quad (r > 0, \varepsilon > 0). \quad (4.2)$$

*The assertion also holds when the derivative cutoff is omitted.*

*Proof.* Expand  $\Xi_i(P)$  into real wedges. Each term is the pullback of real coordinate volume under a map whose first coordinate is  $\Re P$  or  $\Im P$ , and whose other coordinates are one real coordinate from each  $z_j$  with  $j \neq i$ . The image of the indicated set lies in

$$[-\varepsilon, \varepsilon] \times [-L, L]^{d-1}.$$

Apply Lemma 3.4 to the family retaining  $\theta$ , the real and imaginary coefficients of  $P$ ,  $\varepsilon$ , and  $r$ . There are only finitely many choices of  $i$  and real wedge term. Their image volumes are at most  $2\varepsilon(2L)^{d-1}$ . Summing proves the assertion. The triangle inequality is applied only after evaluating each complex form on the tangent vector.  $\square$

*Proof of Theorem 4.1.* We induct on the degree bound  $D$ . Corollary 3.6 gives a bound  $V_0$  for the integral over all of  $E_\theta$ . It suffices to handle  $0 < \varepsilon < 1/2$ ; enlarging the final constant handles  $\varepsilon \geq 1/2$ .

Let  $H = \max\{1, \sqrt{2}L\}$  and

$$B_D = \sum_{0 < |\alpha|_1 \leq D} H^{|\alpha|_1}, \quad c = (2B_D)^{-1}.$$

If every nonconstant coefficient of  $P$  has modulus less than  $c$ , its constant coefficient has modulus 1 and the remaining terms have sum of moduli at most  $1/2$  throughout the box. Thus  $|P| \geq 1/2$  there, and the sublevel set under consideration is empty.

Otherwise choose a nonconstant coefficient  $[z^\beta]P$  of modulus at least  $c$ , and an  $i$  with  $\beta_i > 0$ . Put  $Q = \partial_i P$  and  $M = \|Q\|_{\text{coef}}$ . Distinct monomials remain distinct under this partial differentiation whenever they survive; hence

$$M \geq \beta_i |[z^\beta]P| \geq c. \quad (4.3)$$

Holomorphy gives the exact identity

$$\Xi_i(P) = (-1)^{i-1} Q \Omega_d. \quad (4.4)$$

By (3.2), on  $|Q| \geq r$ ,

$$|\Omega_d| \leq r^{-1} |\Xi_i(P)|.$$

Consequently Lemma 4.2 bounds the large-derivative part by  $C_1 \varepsilon / r$ .

If  $D = 1$ , then  $Q$  is a constant of modulus at least  $c$ . Applying the same argument with  $r = c$  gives the bound  $C_1 \varepsilon / c$ . This establishes the base case, including polynomials of degree zero through the preceding near-constant case.

Suppose  $D \geq 2$ . The induction hypothesis applies to the normalized polynomial  $Q/M$  of degree at most  $D - 1$ , even when it is constant. Discarding the condition  $|P| \leq \varepsilon$  on the small-derivative part, we obtain

$$\begin{aligned} \int_{E_\theta \cap \{|P| \leq \varepsilon\}} |\Omega_d| &\leq C_1 \frac{\varepsilon}{r} + C_{D-1} \left( \frac{r}{M} \right)^{1/(D-1)} \\ &\leq C_1 \frac{\varepsilon}{r} + C_{D-1} c^{-1/(D-1)} r^{1/(D-1)}. \end{aligned} \quad (4.5)$$

Take  $r = \varepsilon^{(D-1)/D}$ . Both terms are constant multiples of  $\varepsilon^{1/D}$ , proving the induction. All constants depend only on the fixed family and degree bounds. Finally, let  $\varepsilon \downarrow 0$  and use monotonicity of the positive integral to see that the integral over  $P = 0$  is zero.  $\square$

*Remark 4.3.* This is a sublevel estimate for a restricted complex top form, not for Euclidean volume of  $E_\theta$ . The distinction permits the estimate when a whole component of  $E_\theta$  lies in  $P = 0$ : on that component the relevant complex form must have zero integral density. No regularity or transversality of the hypersurface  $P = 0$  is assumed.

## 5 Logarithmic integrability on the torus

**Theorem 5.1** (Middle-dimensional logarithmic integrability). *Let  $d \geq 1$ , let  $f \in R_d$  satisfy  $0 \in \text{int Newt}(f)$ , and let  $S \subset (\mathbb{C}^\times)^d$  be semialgebraic of real dimension at most  $d$ . Choose  $b \in \mathbb{Z}_{\geq 0}^d$  with  $b + \text{supp}(f) \subset \mathbb{Z}_{\geq 0}^d$  and put*

$$D = \max_{\alpha \in \text{supp}(f)} |b + \alpha|_1.$$

*Then  $D \geq 1$ , and there is a constant  $C = C(S, f, b)$  such that*

$$\int_{S \cap \{|f| \leq R\}} |\omega_d| \leq C R^{1/D} \quad (R > 0). \quad (5.1)$$

*The same assertion holds for a finite semialgebraic chain, with absolute multiplicities. In particular, the integral of  $|\omega_d|$  is finite on every such chain on which  $f$  is bounded.*

*Proof.* Interiority gives a  $\kappa > 0$  such that  $[-\kappa, \kappa]^d \subset \text{Newt}(f)$ . With  $a_* = \min_{\alpha \in \text{supp}(f)} |a_\alpha| > 0$ , define

$$\rho_k = \max_{\alpha \in \text{supp}(f)} |a_\alpha| 2^{\langle k, \alpha \rangle}, \quad k \in \mathbb{Z}^d.$$

The support function of the contained cube gives

$$\rho_k \geq a_* 2^{\kappa \|k\|_1}. \quad (5.2)$$

Partition the torus into the disjoint half-open boxes

$$B_k = \{x : 2^{k_i} \leq |x_i| < 2^{k_i+1} \ (1 \leq i \leq d)\}.$$

On  $B_k$  write  $x_i = 2^{k_i} z_i$ , so  $1 \leq |z_i| < 2$ . Positive diagonal scaling preserves the logarithmic form. The polynomial

$$P_k(z) = \rho_k^{-1} z^b f(2^{k_1} z_1, \dots, 2^{k_d} z_d)$$

has degree at most  $D$  and coefficient norm exactly 1. There is no combination of distinct exponents in this normalization. If  $|f| \leq R$ , then

$$|P_k(z)| \leq 2^{|b|_1} R / \rho_k. \quad (5.3)$$

The rescaled sets belong to one semialgebraic family

$$E_\lambda = \{z : 1 \leq |z_i| < 2 \text{ for every } i, \ (\lambda_1 z_1, \dots, \lambda_d z_d) \in S\}, \quad \lambda \in (0, \infty)^d.$$

This family lies in a fixed bounded box and has dimension at most  $d$ . It includes every set obtained from  $S \cap B_k$ . On its fixed annulus,

$$\omega_d = (z_1 \cdots z_d)^{-1} \Omega_d, \quad |\omega_d| \leq |\Omega_d|.$$

Applying Theorem 4.1 with this entire family, and then (5.2), gives a constant  $C_0$  independent of  $k$  and  $R$  with

$$\begin{aligned} \int_{S \cap B_k \cap \{|f| \leq R\}} |\omega_d| &\leq C_0 (2^{|b|_1} R / \rho_k)^{1/D} \\ &\leq C_1 R^{1/D} 2^{-\kappa \|k\|_1 / D}. \end{aligned} \quad (5.4)$$

Here  $D = 0$  would force the support of  $f$  to be a singleton, contradicting full-dimensional interiority for  $d \geq 1$ .

Set  $q = 2^{-\kappa/D} \in (0, 1)$ . The convergent sum is explicitly

$$\sum_{k \in \mathbb{Z}^d} q^{\|k\|_1} = \left( 1 + 2 \sum_{j \geq 1} q^j \right)^d = \left( \frac{1+q}{1-q} \right)^d.$$

Countable additivity of the nonnegative integral on the disjoint boxes proves (5.1). Boundary pieces are assigned to exactly one box; they need not have zero measure in  $S$ . Summing the bounds with the finitely many absolute chain multiplicities proves the chain version.  $\square$

## 6 Generalized critical values and ODE transport

We now supply the global transport needed to continue the residue cycle from a large parameter to a punctured interval approaching zero. The point is to do this directly on the affine torus, without compactification, Whitney stratifications, or Thom isotopy.

## 6.1 A proper metric on the affine torus

Write

$$\mathcal{T}_d = \{(z, w) \in \mathbb{C}^{2d} : z_j w_j = 1 \ (1 \leq j \leq d)\}.$$

The map

$$\iota : (\mathbb{C}^\times)^d \longrightarrow \mathcal{T}_d, \quad x \longmapsto (x_1, \dots, x_d, x_1^{-1}, \dots, x_d^{-1})$$

is an algebraic isomorphism onto the closed smooth affine variety  $\mathcal{T}_d$ . Replace every negative power  $x_j^{-m}$  in  $f$  by  $w_j^m$ ; this gives a polynomial  $F$  on  $\mathbb{C}^{2d}$  whose restriction to  $\mathcal{T}_d$  corresponds to  $f$ .

Equip  $\mathcal{T}_d$  with the metric induced by the Euclidean Hermitian metric of  $\mathbb{C}^{2d}$ , and put

$$\rho(p) = \|p\|, \quad p \in \mathcal{T}_d. \quad (6.1)$$

Since  $\mathcal{T}_d$  is closed,  $\rho$  is proper. In the original torus coordinates,

$$\rho(x)^2 = \sum_{j=1}^d (|x_j|^2 + |x_j|^{-2}).$$

Thus approaching a coordinate hyperplane  $x_j = 0$  is correctly counted as escape to infinity.

For  $p \in \mathcal{T}_d$ , let

$$\lambda(p) = \inf_{|u|=1} \|dF_p^* u\|, \quad (6.2)$$

where  $dF_p : T_p \mathcal{T}_d \rightarrow \mathbb{C}$  is regarded as a real linear map and  $dF_p^*$  is its metric adjoint. Because  $F|_{\mathcal{T}_d}$  is holomorphic and the metric is Hermitian, its two real singular values agree; consequently

$$\lambda(p) = \|dF_p\|. \quad (6.3)$$

In particular  $\lambda(p) = 0$  exactly at the critical points of  $F$ .

Define the ordinary and asymptotic critical-value sets by

$$K_0(F) = F(\text{Crit}(F)), \quad (6.4)$$

$$K_\infty(F) = \left\{ c \in \mathbb{C} : \begin{array}{l} \text{there are } p_\nu \in \mathcal{T}_d \text{ with } \rho(p_\nu) \rightarrow \infty, \\ F(p_\nu) \rightarrow c, \quad \rho(p_\nu) \lambda(p_\nu) \rightarrow 0 \end{array} \right\}. \quad (6.5)$$

We henceforth use these definitions through the fixed algebraic identification with  $(\mathbb{C}^\times)^d$ ; the chosen radius is proper on that torus.

**Standard input 6.1** (Semialgebraic control at infinity). For a fixed polynomial function on a closed smooth affine semialgebraic submanifold equipped with an induced semialgebraic metric, the following hold.

- (i) There is an integer  $N \geq 1$  such that an asymptotic critical value can be detected along a sequence satisfying

$$\lambda(p) \leq \rho(p)^{-(1+1/N)}.$$

Equivalently, if

$$\tilde{\Sigma}_R = \{p : \rho(p) \geq R, \lambda(p) \leq \rho(p)^{-(1+1/N)}\},$$

then

$$K_\infty(F) = \bigcap_{R>0} \overline{F(\tilde{\Sigma}_R)}. \quad (6.6)$$

- (ii) For a fixed semialgebraic family of subsets of Euclidean spheres, there is  $M < \infty$  such that any two points in one connected component of a radius- $r$  member can be joined inside that component by a piecewise  $C^1$  curve of length at most  $Mr$ , for all sufficiently large  $r$ .

The first statement is the usual semialgebraic Łojasiewicz inequality at infinity for generalized critical values. The second is the Teissier–Yomdin radial path-length estimate. In the polynomial case these are used together in the direct proof of finiteness of asymptotic critical values in [8, Proposition 4.2]. The Rabier-function formulation for polynomial maps on smooth affine varieties is developed in [9]; see also the quantitative generalized Bertini–Sard theorem [10]. We use only the two explicit consequences stated above.

**Proposition 6.2** (Finiteness of generalized critical values). *For nonconstant  $f \in R_d$ , the set*

$$B := K_0(F) \cup K_\infty(F) \subset \mathbb{C}$$

*is finite.*

*Proof.* The critical locus of the polynomial  $F|_{\tau_d}$  is algebraic. It has finitely many irreducible components, and  $F$  is constant on each: on the smooth dense locus of such a component its differential vanishes. Hence  $K_0(F)$  is finite.

It remains to treat  $K_\infty(F)$ . Fix  $N$  from Standard input 6.1, and put

$$\Sigma_r = \tilde{\Sigma}_{R_0} \cap \{\rho = r\}.$$

Apply Hardt triviality from Standard input 3.1 to the radius map on  $\tilde{\Sigma}_{R_0}$  and increase  $R_0$  past the last exceptional radius. The unbounded tail then has finitely many connected components  $E_1, \dots, E_k$ , each meeting every sphere  $\{\rho = r\}$ ,  $r \geq R_0$ , in a connected component  $L_{j,r}$  of  $\Sigma_r$ .

For each  $j$ , semialgebraic selection and finite decomposition give a  $C^1$  semialgebraic section

$$\gamma_j : [R_0, \infty) \rightarrow E_j, \quad \rho(\gamma_j(r)) = r.$$

After increasing  $R_0$  again,  $\|\gamma'_j(r)\| \leq 2$ . Indeed,  $u_j(r) = \gamma_j(r)/r$  is a bounded semialgebraic curve on the unit sphere. By semialgebraic monotonicity (equivalently, its Puiseux behavior at infinity),  $u_j(r)$  has a limit and  $ru'_j(r) \rightarrow 0$ ; thus  $\gamma'_j(r) = u_j(r) + ru'_j(r)$  has norm tending to 1. Using (6.3),

$$|(F \circ \gamma_j)'(r)| \leq \lambda(\gamma_j(r)) \|\gamma'_j(r)\| \leq 2r^{-(1+1/N)}.$$

Therefore  $F(\gamma_j(r))$  converges to a limit  $c_j \in \mathbb{C}$ , with

$$|F(\gamma_j(r)) - c_j| \leq 2Nr^{-1/N}. \tag{6.7}$$

By Standard input 6.1(ii), any  $p \in L_{j,r}$  can be joined to  $\gamma_j(r)$  inside  $L_{j,r}$  by a curve of length at most  $Mr$ . Along that curve  $\|dF\| = \lambda \leq r^{-(1+1/N)}$ , so

$$|F(p) - F(\gamma_j(r))| \leq Mr^{-1/N}. \tag{6.8}$$

Combining (6.7) and (6.8), the image of  $E_j \cap \{\rho \geq R\}$  is contained in a ball about  $c_j$  whose radius is  $O(R^{-1/N})$ . Consequently every sequence in  $\tilde{\Sigma}_R$  escaping to infinity and having bounded  $F$ -value has, after passage to one of the finitely many components, limit  $c_j$ . Equation (6.6) gives

$$K_\infty(F) \subset \{c_1, \dots, c_k\}.$$

Thus  $B$  is finite. □

**Proposition 6.3** (Complete ODE transport). *Let  $B$  be the finite set in Proposition 6.2 and put  $U = \mathbb{C} \setminus B$ . Every fiber over  $U$  is smooth. Moreover, for every piecewise  $C^1$  path  $\alpha : [0, 1] \rightarrow U$ , the minimal-norm horizontal ODE gives a complete transport diffeomorphism*

$$X_{\alpha(0)} \simeq X_{\alpha(1)}.$$

*In particular*

$$f : f^{-1}(U) \longrightarrow U$$

*is locally smoothly trivial, and its fiber homology groups form a local system on  $U$ .*

*Proof.* Let  $Q = \alpha([0, 1])$ . Since  $Q$  is compact and disjoint from  $K_0(F) \cup K_\infty(F)$ , there is  $c_Q > 0$  such that

$$(1 + \rho(p))\lambda(p) \geq c_Q \quad (F(p) \in Q). \quad (6.9)$$

Otherwise there would be a sequence  $p_\nu$  with  $F(p_\nu) \in Q$  and  $(1 + \rho(p_\nu))\lambda(p_\nu) \rightarrow 0$ . If  $\rho(p_\nu)$  were bounded, properness of  $\rho$  would give a convergent subsequence ending at a critical point with value in  $Q$ , contrary to  $Q \cap K_0(F) = \emptyset$ . If it were unbounded, pass to a subsequence with  $\rho(p_\nu) \rightarrow \infty$  and  $F(p_\nu) \rightarrow c \in Q$ ; then  $c \in K_\infty(F)$ , again a contradiction.

For a regular point  $p$ , set  $A_p = dF_p$  and let

$$R_p = A_p^*(A_p A_p^*)^{-1} : \mathbb{C} \rightarrow T_p \mathcal{T}_d$$

be its minimal-norm real right inverse. It depends smoothly on  $p$  and satisfies

$$A_p R_p = \text{id}_{\mathbb{C}}, \quad \|R_p\| = \lambda(p)^{-1}.$$

Along a  $C^1$  piece of  $\alpha$ , consider

$$p'(t) = R_{p(t)} \alpha'(t), \quad F(p(0)) = \alpha(0). \quad (6.10)$$

Then

$$\frac{d}{dt} F(p(t)) = \alpha'(t),$$

so  $F(p(t)) = \alpha(t)$  while the solution exists. By (6.9),

$$\|p'(t)\| \leq C_\alpha (1 + \rho(p(t))) \quad (6.11)$$

for a constant depending only on the path. Since  $|\rho'(t)| \leq \|p'(t)\|$ , Gronwall's inequality bounds  $\rho(p(t))$  on every finite time interval. The sublevel sets of  $\rho$  are compact in  $\mathcal{T}_d$ , so the ordinary ODE continuation theorem extends the solution across all of  $[0, 1]$ . This excludes both ordinary Euclidean escape and approach to a deleted coordinate hyperplane.

Transport along the reversed path gives the inverse map, and smooth dependence on initial conditions gives a diffeomorphism of fibers. Applying the same construction to the straight radial paths in a small disc compactly contained in  $U$  yields a smooth local trivialization. The homology groups therefore form a local system.  $\square$

## 6.2 Relative forms and holomorphic periods

At a regular value  $s$ , put  $X_s = f^{-1}(s)$ . There is a canonical holomorphic  $(d-1)$ -form  $\eta_s$  on  $X_s$ , denoted  $\omega_d/df$ : locally choose a  $(d-1)$ -form  $\eta$  with  $df \wedge \eta = \omega_d$  and restrict it to the fiber. Differences between two such choices restrict to zero. In local holomorphic coordinates  $(f, z'_2, \dots, z'_d)$ , this is the ordinary relative holomorphic top form. In particular  $d\eta_s = 0$  on the fiber.

**Lemma 6.4** (Holomorphy of the transported period). *Let  $[\gamma_s] \in H_{d-1}(X_s; \mathbb{Z})$  be a locally constant homology class over a simply connected open subset of  $U$  from Proposition 6.3. Then*

$$J(s) = \int_{\gamma_s} \eta_s$$

*is well defined and holomorphic. Consequently its germ can be continued along any path in  $U$ .*

*Proof.* Closedness of  $\eta_s$  and the ordinary de Rham pairing show that the integral depends only on the homology class. Proposition 6.3 provides smooth local trivializations by complete horizontal ODE transport. Choose a compact smooth singular cycle in one fiber and transport it in a small disc; all transported cycles remain in the image of one compact cycle under a smooth map from a compact parameter set.

For a small oriented triangle  $\Delta$  in the disc, use such a local trivialization to sweep the cycle over  $\Delta$ . This is a  $(d+1)$ -chain  $\Gamma_\Delta$ , whose boundary is the sweep over  $\partial\Delta$ . On each edge,  $\omega_d = df \wedge \eta$  and  $f$  is the base coordinate, so ordinary parametrized integration gives

$$\int_{\partial\Delta} J(s) ds = \int_{\partial\Gamma_\Delta} \omega_d = \int_{\Gamma_\Delta} d\omega_d = 0.$$

The holomorphic top form  $\omega_d$  is closed. Continuity of  $J$  follows from the same compact smooth family, and Morera's theorem proves holomorphy. Repeating this along overlapping discs gives continuation along every path in  $U$ . No global monodromy invariance of the transported class is required.  $\square$

## 7 Semialgebraic sweeps and the endpoint primitive

Continue to assume  $0 \in \text{int Newt}(f)$  and  $d \geq 1$ . Choose  $b > 0$  sufficiently small that  $(0, b] \subset U$ . Applying Hardt triviality to the restriction of  $f$  over the real interval, and decreasing  $b$  once more if necessary, gives a semialgebraic homeomorphism

$$\Psi : (0, b] \times X_b \longrightarrow f^{-1}((0, b]), \quad f(\Psi(s, z)) = s, \quad \Psi(b, z) = z. \quad (7.1)$$

The last normalization is obtained by composing the fiber coordinate with the inverse of its value at  $b$ .

Fix a compact semialgebraic cycle  $\gamma_b$  representing the prescribed transported class at  $b$ , and put  $\gamma_s = \Psi(s, \gamma_b)$ . These classes coincide with those used for analytic continuation. Indeed, both are locally constant in the same fiber-homology local system and agree at  $b$  on the connected interval. Thus their periods are the function  $J$  of Lemma 6.4.

Triangulate the finite cycle  $\gamma_b$  and sweep its simplices using (7.1). Denote the resulting finite semialgebraic  $d$ -chain by  $S$ , and its restriction over  $[a, c] \subset (0, b]$  by  $S_{[a, c]}$ . Orient the base first, so

$$\partial S_{[a, c]} = \gamma_c - \gamma_a. \quad (7.2)$$

Shared simplex faces cancel with their cycle multiplicities. Each truncated sweep is compact: it is the continuous image of a finite compact cycle times  $[a, c]$ . Its support lies in the regular locus of  $f$ . The full sweep need not have compact closure in the torus.

**Proposition 7.1** (Quantitative endpoint primitive). *There are  $C > 0$ ,  $\delta > 0$ , and a function  $I : (0, b] \rightarrow \mathbb{C}$ , continuous on  $(0, b]$  and  $C^1$  on  $(0, b)$ , such that*

$$I'(s) = J(s), \quad (7.3)$$

*and  $I$  extends continuously to 0 with*

$$|I(t) - I(0)| \leq Ct^\delta \quad (0 < t \leq b). \quad (7.4)$$

*In particular, for  $0 < a < c \leq b$ ,*

$$\int_a^c J(s) \, ds = \int_{S_{[a,c]}} \omega_d, \quad (7.5)$$

*and, after an enlargement of  $C$  if needed,*

$$\left| \int_a^{2a} J(s) \, ds \right| \leq C(2a)^\delta \quad (0 < 2a \leq b). \quad (7.6)$$

*Proof.* By Theorem 5.1, with  $\delta = 1/D$  as there,

$$\int_{|S| \cap \{|f| \leq r\}} |\omega_d| \leq Cr^\delta \quad (r > 0). \quad (7.7)$$

This applies separately to the finitely many swept simplices and then with their absolute multiplicities. In particular the integral over the whole sweep is absolutely convergent, since  $|f| \leq b$  there. Define

$$I(t) = - \int_{S_{[t,b]}} \omega_d, \quad I(0) = - \int_S \omega_d.$$

The possible endpoint slices have dimension at most  $d - 1$  and contribute zero to these  $d$ -integrals. Since  $f = s$  on the sweep, (7.7) proves (7.4). It remains to prove (7.3); the fundamental theorem of calculus will then give (7.5).

On the regular locus of  $f$ , define the smooth complex vector field and smooth  $(d - 1)$ -form

$$V = \frac{\sum_{j=1}^d \overline{\partial_j f} \, \partial / \partial x_j}{\sum_{j=1}^d |\partial_j f|^2}, \quad \eta = \iota_V \omega_d. \quad (7.8)$$

Then  $df(V) = 1$  and  $df \wedge \eta = \omega_d$ ; the latter follows by contracting the identically zero  $(d + 1)$ -form  $df \wedge \omega_d$ . The restriction of  $\eta$  to  $X_s$  is  $\eta_s$ .

For  $0 < u < v \leq b$ , the identity

$$\omega_d = d((f - u)\eta) - (f - u)d\eta$$

and Stokes on the compact semialgebraic sweep give

$$I(v) - I(u) = (v - u)J(v) - \int_{S_{[u,v]}} (f - u)d\eta. \quad (7.9)$$

The lower boundary term is zero, and the upper one is  $(v - u) \int_{\gamma_v} \eta_v$ . This is the local differentiation identity in [3, Lemma 9].

Fix  $s_0 \in (0, b)$ . All sweeps over a sufficiently small closed neighborhood of  $s_0$  lie in one compact subset of the regular locus. There  $d\eta$  has bounded norm. Since  $f$  takes values in  $[u, v]$  on  $S_{[u,v]}$ , the error in the difference quotient is bounded by

$$\left| \frac{I(v) - I(u)}{v - u} - J(v) \right| \leq C_{s_0} \text{mass}_d(S_{[u,v]}). \quad (7.10)$$

Here mass means the sum of the Euclidean  $d$ -volumes of the fixed swept pieces with their absolute multiplicities. It is finite on each fixed compact truncation by Corollary 3.6. As  $v \downarrow u = s_0$ , the sets decrease to a  $(d-1)$ -dimensional slice, so this mass tends to zero by continuity from above of a finite measure. The same argument applies from the other side. Continuity of  $J$ , proved independently in Lemma 6.4, yields  $I'(s_0) = J(s_0)$ . Thus  $I$  is  $C^1$  on  $(0, b)$ . The same compact-truncation argument at the upper endpoint gives  $I(t) \rightarrow I(b) = 0$  as  $t \uparrow b$ .

Finally, (7.5) and (7.7), applied to  $S_{[a, 2a]} \subset \{|f| \leq 2a\}$ , prove (7.6). No limiting Stokes theorem at  $s = 0$  has been used: Stokes was applied only on compact truncations away from zero.  $\square$

**Corollary 7.2** (Exclusion of the forced simple pole). *The period in Proposition 7.1 cannot satisfy  $J(s) = A/s$  on  $(0, b]$  for any constant  $A \neq 0$ .*

*Proof.* For  $0 < 2a \leq b$ ,

$$\int_a^{2a} \frac{A}{s} ds = A \log 2.$$

Its modulus is the positive constant  $|A| \log 2$ , whereas (7.6) tends to zero as  $a \downarrow 0$ .  $\square$

## 8 The local residue identity and minimal nonvanishing

By Lemmas 2.1 and 2.3, the only case still requiring proof is  $d \geq 1$ ,  $0 \in \text{int Newt}(f)$ , and

$$f(y) = y^{-m} u(y), \quad m_i \geq 1, \quad u(0) \neq 0.$$

All constant terms are computed in these transformed variables, where they equal the original constant terms. Set

$$\omega = \frac{dy_1}{y_1} \wedge \cdots \wedge \frac{dy_d}{y_d}.$$

**Lemma 8.1** (Residue cycles near a vertex). *For all sufficiently large  $|s|$ , there is a compact semialgebraic  $(d-1)$ -cycle  $\gamma_s \subset f^{-1}(s)$ , varying locally smoothly, such that*

$$R(s) := \sum_{n \geq 0} \frac{\text{CT}(f^n)}{s^{n+1}} = -(2\pi i)^{1-d} \int_{\gamma_s} \frac{\omega}{df}. \quad (8.1)$$

*The series converges for all sufficiently large  $|s|$ .*

*Proof.* Choose  $\varepsilon > 0$  so small that  $u$  has no zero on the closed polydisc  $|y_i| \leq \varepsilon$  and

$$|y_1 \partial_1 u| < m_1 |u| \quad \text{on that polydisc.} \quad (8.2)$$

Let  $K_\varepsilon = \{|y_i| = \varepsilon \ (1 \leq i \leq d)\}$  with its ordered product-circle orientation. If  $|s| > \max_{K_\varepsilon} |f|$ , the geometric series is uniformly convergent on  $K_\varepsilon$  and coefficient extraction gives

$$R(s) = \frac{1}{(2\pi i)^d} \int_{K_\varepsilon} \frac{\omega}{s - f}. \quad (8.3)$$

Indeed the integral of each Laurent monomial is zero unless its exponent is zero, in which case it equals  $(2\pi i)^d$ .

For  $y' = (y_2, \dots, y_d)$  on the product circle  $K' = \{|y_j| = \varepsilon \ (j > 1)\}$ , solve

$$s y_1^{m_1} \prod_{j>1} y_j^{m_j} - u(y_1, y') = 0. \quad (8.4)$$

For  $|s|$  uniformly large, Rouché's theorem on  $|y_1| = \varepsilon$  shows that there are exactly  $m_1$  roots in the disc, counted with multiplicity. None is zero because  $u(0, y') \neq 0$ . At a root, the  $y_1$ -derivative of the left side of (8.4) is  $(m_1 u - y_1 \partial_1 u)/y_1$ , which is nonzero by (8.2). Thus all roots are simple.

Let  $\gamma_s$  be the union of these roots over  $K'$ . It is a compact unbranched  $m_1$ -sheeted cover of  $K'$ , oriented by the local projections onto the ordered base circles. Compactness follows uniformly from the absence of roots on either  $y_1 = 0$  or  $|y_1| = \varepsilon$  and compactness of  $K'$ . The defining equations and inequalities are semialgebraic. The cover has no boundary and so defines a cycle. For  $d = 1$ , interpret  $K'$  as a point and  $\gamma_s$  as the positively weighted finite set of roots.

As a one-variable form in  $y_1$ ,

$$\frac{dy_1}{y_1(s - f)} = \frac{y_1^{m_1-1} (\prod_{j>1} y_j^{m_j}) dy_1}{s y_1^{m_1} \prod_{j>1} y_j^{m_j} - u(y)}.$$

It is holomorphic at  $y_1 = 0$ . At a root  $\zeta$  its residue is  $-1/(\zeta \partial_1 f(\zeta, y'))$ . On this part of the fiber,

$$\frac{\omega}{df} = \frac{1}{y_1 \partial_1 f} \frac{dy_2}{y_2} \wedge \cdots \wedge \frac{dy_d}{y_d}.$$

Integrating first in  $y_1$  and then over  $K'$  gives

$$\int_{K_\varepsilon} \frac{\omega}{s - f} = -2\pi i \int_{\gamma_s} \frac{\omega}{df}.$$

Together with (8.3), this proves (8.1), including its sign and normalization. The implicit function theorem at the simple roots gives local smooth variation of the cycle and the associated locally constant homology class.  $\square$

*Proof of Theorem 1.2.* If the minimal-face reduction has rank zero,  $f_Q$  is a nonzero constant and its first constant term is nonzero. Otherwise apply the reduction and unimodular change just described. Suppose for contradiction that  $\text{CT}(f^n) = 0$  for every  $n \geq 1$ . Lemma 8.1 then gives

$$R(s) = \frac{1}{s}, \quad J(s) := \int_{\gamma_s} \frac{\omega}{df} = -\frac{(2\pi i)^{d-1}}{s}$$

for all sufficiently large  $|s|$ .

Let  $B$  be the finite set in Proposition 6.3. Choose a large base point outside  $B \cup \{0\}$  and continue the homology class of the residue cycle along a path in  $\mathbb{C} \setminus (B \cup \{0\})$  to a sufficiently small positive  $b$ . Such a path exists because the complement of finitely many points in  $\mathbb{C}$  is path connected. Lemma 6.4 continues the scalar period. Since the rational function  $-(2\pi i)^{d-1}/s$  is holomorphic along the same path, uniqueness of analytic continuation gives

$$J(s) = -\frac{(2\pi i)^{d-1}}{s} \quad (0 < s \leq b).$$

Here  $J$  denotes the period of the continued homology class, not the integral over the original fixed product torus after leaving its initial convergence region. The cycle may have monodromy; the displayed scalar identity is what is continued.

Proposition 7.1 applies to this class and its semialgebraic representatives on  $(0, b]$ . Corollary 7.2 contradicts the displayed identity, since  $(2\pi i)^{d-1} \neq 0$ . The assumed universal vanishing is therefore impossible.  $\square$

## 9 Infinite nonvanishing and the torus Mathieu property

*Proof of Theorem 1.1.* Fix  $M \geq 0$  and set  $q = M + 1$ . By Lemma 2.2,  $0 \in \text{Newt}(f^q)$ . Theorem 1.2 gives  $k \geq 1$  with

$$0 \neq \text{CT}((f^q)^k) = \text{CT}(f^{qk}).$$

Since  $qk \geq q > M$ , this proves the assertion. In particular, no separate argument concerning higher-order poles is required.  $\square$

**Corollary 9.1** (Newton-polytope classification). *For  $0 \neq f \in R_d$ ,*

$$\text{CT}(f^n) = 0 \text{ for every } n \geq 1 \iff 0 \notin \text{Newt}(f).$$

*Proof.* One direction is the contrapositive of Theorem 1.2. For the other, strict separation of a point and a compact convex set gives a linear functional  $\ell$  and  $\delta > 0$  with  $\ell(\alpha) \geq \delta$  for every  $\alpha \in \text{supp}(f)$ . Every exponent in  $f^n$  is a sum of  $n$  such exponents, and has  $\ell$ -value at least  $n\delta > 0$ . The zero exponent cannot occur.  $\square$

**Corollary 9.2** (Laurent-polynomial Mathieu property). *Let  $f, h \in R_d$ . If  $\text{CT}(f^n) = 0$  for every  $n \geq 1$ , then  $\text{CT}(hf^n) = 0$  for all sufficiently large  $n$ .*

*Proof.* The assertion is immediate if  $f = 0$  or  $h = 0$ . Otherwise use Corollary 9.1 and choose  $\ell, \delta$  as in its proof. Put  $A_h = \min_{\beta \in \text{supp}(h)} \ell(\beta)$ . Any exponent in  $hf^n$  has  $\ell$ -value at least  $A_h + n\delta$ , which is positive for all sufficiently large  $n$ . Thus its constant coefficient is zero.  $\square$

**Corollary 9.3** (Compact tori). *Let  $K = \mathbb{T}^d$  with normalized Haar measure. For finite Fourier sums  $F, H$  on  $K$ , if  $\int_K F^n = 0$  for every  $n \geq 1$ , then  $\int_K HF^n = 0$  for all sufficiently large  $n$ .*

*Proof.* Identify a character with  $x^\alpha = e^{i\langle \alpha, \theta \rangle}$ ,  $\alpha \in \mathbb{Z}^d$ . Direct integration in each circle coordinate gives

$$\frac{1}{(2\pi)^d} \int_{[0, 2\pi]^d} e^{i\langle \alpha, \theta \rangle} d\theta = \begin{cases} 1, & \alpha = 0, \\ 0, & \alpha \neq 0. \end{cases}$$

Hence the Haar integral of any Laurent polynomial restricted to  $K$  is its constant coefficient. Apply Corollary 9.2.  $\square$

*Remark 9.4.* Finite Fourier sums are exactly the continuous  $K$ -finite functions on a torus. One direction follows because each character spans a one-dimensional translation module. Conversely, on a finite-dimensional translation-invariant space, average an inner product over  $K$  and simultaneously diagonalize the commuting unitary translations. An eigenfunction is a scalar multiple of the corresponding character, by evaluation at the identity. The continuous characters of  $\mathbb{T}^d$  are precisely those indexed by  $\mathbb{Z}^d$ . This is the usual torus interpretation of [3]; no assertion about nonabelian groups is needed in this paper.

## 10 Provenance and mathematical scope

The theorem proved is the arbitrary-rank nonvanishing theorem of Duistermaat and van der Kallen. The residue construction in Lemma 8.1 is their local construction, and the calculation (7.9) is their primitive-differentiation identity. The contribution proposed for independent examination is the uniform middle-dimensional sublevel estimate and its dyadic use in Theorem 5.1. This replaces the argument making the Haar form regular over finite fibers of a smooth compactification.

The semialgebraic integration estimates in [6] are antecedents, not conclusions attributed to the present paper. The coefficient-derivative induction in Section 4 is also naturally compared with geometric sublevel-set estimates; see, for example, [5] and the references there. The specific estimate here concerns the modulus of a restricted holomorphic top form and is uniform in a fixed semialgebraic family. These precise hypotheses, rather than merely similar terminology, must be used in any priority comparison.

Question 9.1 of [12] motivates the search for a simpler nonvanishing proof. The present route retains substantial, explicitly identified semialgebraic inputs: Hardt triviality, the Łojasiewicz control of generalized critical values at infinity, and a uniform radial path-length estimate. The normalized-gradient/right-inverse transport in Section 6 is the standard Malgrange–Rabier mechanism in a proper metric on the affine torus; compare [8, 9, 10]. No Whitney stratification or Thom isotopy is used in the main proof, although Appendix A retains that earlier route for comparison. The argument is therefore described as resolution-free rather than as elementary in every respect. The stronger singularity and growth conclusions of [3, Theorem 5] are outside the claims made here. A proof and generalization using Newton–Puiseux methods in one variable is discussed in [4]; the present theorem has no rank restriction.

The intended future formalization has one core endpoint, Theorem 1.2. Lemma 2.2 and the separation argument then give Theorem 1.1 and Corollary 9.2. Standard inputs in Section 3 and Section 6 remain genuine mathematical dependencies: a formal proof must establish them or invoke proved library theorems, not add them as unproved project axioms. The main formalization target is the generalized-critical-value/ODE route; the Whitney–Thom appendix is an alternate proof, not a required dependency.

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## A Whitney–Thom alternative for finite bifurcation and cycle transport

This appendix records the compactification-and-stratification route that was used in an earlier version of the proof. It is logically independent of the generalized-critical-value/ODE transport in Section 6 and is not used in the main argument.

**Standard input A.1** (Complex algebraic stratification and first isotopy). A complex algebraic variety, together with finitely many specified algebraic subsets, admits a finite complex algebraic Whitney stratification compatible with those subsets. If a map from a closed Whitney-stratified subset of a smooth manifold to a smooth base is proper and restricts to a submersion on every stratum, it is locally topologically trivial by homeomorphisms preserving the strata.

These are the algebraic Whitney-stratification theorem and Thom’s first isotopy lemma; see [11, 13]. For a direct treatment of nonproper polynomial maps, see [2], especially Theorems 3.1 and 5.1 of its arXiv version.

**Proposition A.2** (Whitney–Thom finite exceptional set). *For a nonconstant  $f \in R_d$ , there is a finite set  $B_W \subset \mathbb{C}$  such that all fibers over  $U_W = \mathbb{C} \setminus B_W$  are smooth and*

$$f : f^{-1}(U_W) \longrightarrow U_W$$

is locally topologically trivial.

*Proof.* Let  $\bar{\Gamma}$  be the closure of the graph of  $f$  in  $(\mathbb{P}^1)^d \times \mathbb{P}^1$ , let  $p : \bar{\Gamma} \rightarrow \mathbb{P}^1$  be projection, and let  $D_\infty$  be the complement of the open graph identified with  $(\mathbb{C}^\times)^d$ . The projection  $p$  is proper. Choose a finite complex algebraic Whitney stratification compatible with  $D_\infty$ .

For a stratum  $W$ , the critical locus of  $p|_W$  is algebraic in  $W$ . At such a point its complex differential has rank zero, because the base has complex dimension one. On the smooth locus of any irreducible component of this critical locus, the restriction of  $p$  consequently has zero differential and is locally constant. It is therefore constant on that component. Equivalently, a rational function with zero differential on an irreducible complex variety is constant. There are only finitely many components and strata, so their critical images form a finite subset of  $\mathbb{P}^1$ .

Remove that finite subset and the value  $\infty$ . On the remaining base,  $p$  is a proper stratified submersion, so Standard input A.1 gives local trivializations preserving all strata. They preserve  $D_\infty$  and its complement. Restricting to the latter gives the claimed local trivializations of  $f$ . At a point of this open graph, submersivity on its stratum implies submersivity on the ambient smooth torus; hence every remaining fiber is smooth. Empty fibers, if present, cause no difficulty in this statement. In the application a nonempty fiber is supplied by the explicit residue construction.  $\square$

*Remark A.3.* The adjective *complex algebraic* matters in this appendix. Real semialgebraic strata can map to curves of real critical values. Such a real exceptional set need not leave a connected complement through which the required analytic germ can be transported. The projective graph may be singular; no resolution theorem is hidden in this argument.

*Remark A.4.* Proposition A.2 can replace Propositions 6.2–6.3 in the nonvanishing proof. For the period itself, one then uses the local compact-cycle argument from the earlier draft: on a sufficiently small disc of regular values, lift the two real coordinate vector fields only near the compact cycle support, transport the finite cycle smoothly, and apply the triangle-sweep/Morera argument of Lemma 6.4. Thus the appendix route supplies the same analytic continuation without any appeal to the ODE completeness argument of Section 6.

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